8th Grade Physical Science - Work & Simple Machines

Name: _____ Date: _____

Conservation of energy: an ideal machine has equal input and output work. It trades force for distance without creating energy. Assume no friction, negligible machine mass, and slow, steady motion. Real friction turns some input energy into heat; total energy is conserved.

1 = input; 2 = output. F is force (N); d is distance in the force direction (m). For levers, use vertical rise/fall. Work is measured in joules (J).

$\mathbf{F_1 d_1 = F_2 d_2}$ $\quad MA = F_2 / F_1 = d_1 / d_2$ $\quad F_1 = (F_2 \times d_2) \div d_1$

Example: $F_1 = (160 \times 0.30) \div 1.20 = 40$ N. Both works = 48 J; MA = 4. The input moves 4 times as far, so the input force is 1/4 as large.

Each problem: use $F_1 d_1 = F_2 d_2$, find the missing value and MA, then compare distance and force.

1. Equal-arm lever

Find F_1 for this equal-arm lever. The input falls 0.24 m as the 72 N load rises 0.24 m.

$F_1 d_1 = F_2 d_2$: _____

F_1 = _____ MA = _____

Farther: input / output / neither

Input force: smaller / equal / larger than output

2. A longer lever arm

Find F_1. The long input arm falls 0.60 m as the 320 N load rises 0.15 m.

$F_1 d_1 = F_2 d_2$: _____

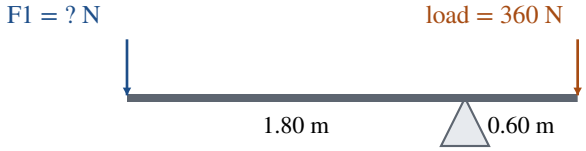
F_1 = _____ MA = _____

Farther: input / output / neither

Input force: smaller / equal / larger than output

3. Balance a seesaw

Find F_1 to balance the 360 N weight. Use the slow-tilt distances below in $F_1 d_1 = F_2 d_2$.



slow tilt: left falls 0.45 m; right rises 0.15 m

$F_1 d_1 = F_2 d_2$: _____

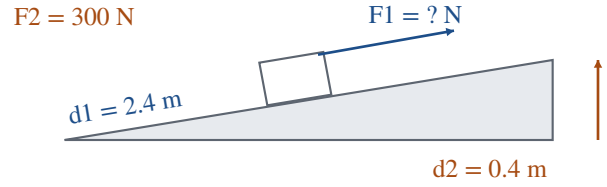
F_1 = _____ MA = _____

Farther: input / output / neither

Input force: smaller / equal / larger than output

4. Inclined plane

Find the push F_1 parallel to the ramp. The 300 N load moves 2.40 m along it and rises 0.40 m.



$F_1 d_1 = F_2 d_2$: _____

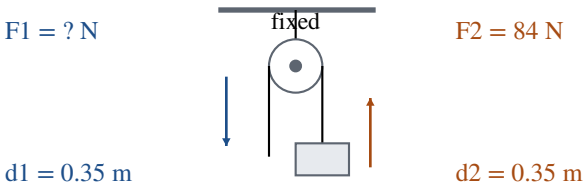
F_1 = _____ MA = _____

Farther: input / output / neither

Input force: smaller / equal / larger than output

5. Fixed pulley

Find F_1. Pull the rope down 0.35 m to lift an 84 N load by 0.35 m.



$F_1 d_1 = F_2 d_2$: _____

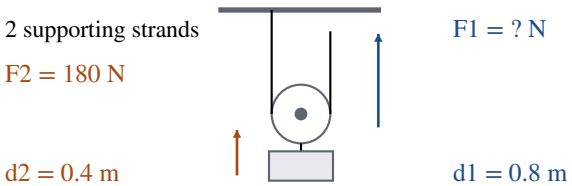
F_1 = _____ MA = _____

Farther: input / output / neither

Input force: smaller / equal / larger than output

6. Movable pulley

Find F_1. Pull 0.80 m of rope upward to lift the 180 N load by 0.40 m.



$F_1 d_1 = F_2 d_2$: _____

F_1 = _____ MA = _____

Farther: input / output / neither

Input force: smaller / equal / larger than output

Seesaw #3: at rest, work = 0. Balance: F1 × 1.80 m = 360 N × 0.60 m.

Optional app: GabrielW1005.github.io/machine-lab/

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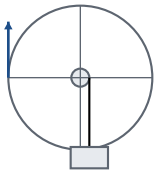
For every problem: $F_1 d_1 = F_2 d_2$; $MA = F_2 / F_1 = d_1 / d_2$. Find the missing value and MA. Which travels farther: input, output, or neither? Is the input force smaller, equal, or larger?

The six simple machines are lever, inclined plane, pulley, wheel and axle, screw, and wedge. The geared winch is a compound machine. Linked syringes are a related hydraulic machine.

7. Wheel and axle

Find F_1 at the wheel rim. Your hand travels 0.96 m while the axle lifts a 240 N load by 0.12 m.

F1 = ? N



F2 = 240 N

rim path

d1 = 0.96 m

rope rise

d2 = 0.12 m

$F_1 d_1 = F_2 d_2$: _____

F_1 = _____ MA = _____

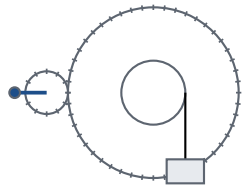
Farther: input / output / neither

Input force: smaller / equal / larger than output

8. Geared winch

Find F_2. A 24 N input turns the crank through 0.80 m of path; the rope rises 0.20 m. Crank and drum radii are equal.

F1 = 24 N



F2 = ? N

d1 = 0.8 m

d2 = 0.2 m

$F_1 d_1 = F_2 d_2$: _____

F_2 = _____ MA = _____

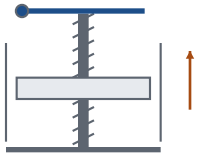
Farther: input / output / neither

Input force: smaller / equal / larger than output

9. Screw lift

Find F_1. One handle turn travels 0.24 m and raises a guided 600 N platform by 0.006 m.

F1 = ? N



F2 = 600 N

one full turn

d1 = 0.24 m

platform rise

d2 = 0.006 m

$F_1 d_1 = F_2 d_2$: _____

F_1 = _____ MA = _____

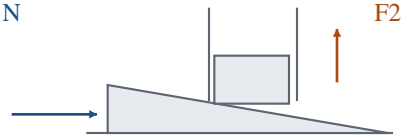
Farther: input / output / neither

Input force: smaller / equal / larger than output

10. A moving wedge

Find d_1, the horizontal push distance. A 65 N push raises the guided 390 N load by 0.020 m.

F1 = 65 N



F2 = 390 N

d1 = ? m

d2 = 0.02 m

$F_1 d_1 = F_2 d_2$: _____

d_1 = _____ MA = _____

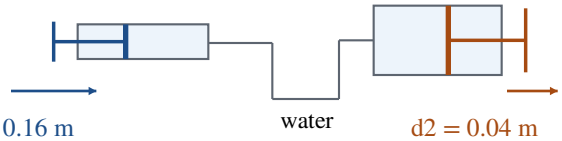
Farther: input / output / neither

Input force: smaller / equal / larger than output

11. Linked syringes

Find F_2. Push the small plunger with 18 N through 0.16 m. The large plunger moves 0.040 m.

F1 = 18 N



F2 = ? N

d1 = 0.16 m

d2 = 0.04 m

$F_1 d_1 = F_2 d_2$: _____

F_2 = _____ MA = _____

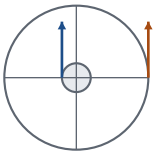
Farther: input / output / neither

Input force: smaller / equal / larger than output

12. Run it in reverse

Find d_2. Drive the small axle with 200 N through 0.080 m. The larger wheel produces 40 N.

F1 = 200 N



F2 = 40 N

axle rim path

d1 = 0.08 m

wheel rim path

d2 = ? m

$F_1 d_1 = F_2 d_2$: _____

d_2 = _____ MA = _____

Farther: input / output / neither

Input force: smaller / equal / larger than output

MA > 1: force advantage. MA = 1: equal forces. MA < 1: output moves farther with less force.

Optional app: GabrielW1005.github.io/machine-lab/